

### Task 1: research

A workflow is a system for managing repetitive processes and tasks which occur in a particular order. They are the mechanism by which people and enterprises accomplish their work, whether manufacturing a product, providing a service, processing information or any other value-generating activity. {<https://www.ibm.com/think/topics/workflow>}

Research into platforms like Camunda reveals a strict separation between a static workflow definition and an active running instance. A definition acts as a reusable blueprint, while an instance holds dynamic runtime data.

The system cannot hard-code workflow structures. The architecture must allow the generic engine to systematically construct different workflow variations from a common set of parts or clone a static blueprint to spawn an independent instance. This ensures multiple running workflows do not accidentally overwrite each other's data.

A workflow management system controls the workflows in a company. Its about task management and it comprises of at least 3 key features and functions{<https://easy-software.com/en/workflow-management-system/#workflow-management-system>} :

- Modeling and simulation: a graphical representation of workflows consisting of how many work steps for the sake of easier handling. Workflows first have to be entered, divided into subsections, precisely described and finalised before they can be displayed in the workflow management system.
- Instantiation and execution: the workflow management system must instantiate exact and case-specific processes so that it will later be able to execute the workflow models. Instantiate means that the workflow management system has an idea of which people, data, applications and such that it can interact with via the workflow model. In practice, this can mean that the workflow management system determines the appropriate processors and programs for all sub-steps. Once this has been done, the workflow must be executed within the relevant programs, initiated by the workflow management system.
- Monitoring and analysis: the monitoring of current work steps and the analysis of processes that have already been carried out. This helps to identify errors. The workflow management system also uses this data to generate meaningful reports and statistics with the goal of improving workflows within the company.

Real-world systems must model nested activities and sub-processes. Platforms like Asana allow individual work items to contain complex, nested sub-tasks, while n8n allows a node to trigger an entirely separate sub-workflow. The core engine should not require separate execution logic for a single task versus a massive sub-process. TIVIDY must define a common interface for all work items so the engine can execute hierarchical structures uniformly, keeping internal storage structures hidden from the engine.

{<https://www.activepieces.com/resources/glossary/nested-flows>}

Workflow engines remain generic by making routing decisions configurable rather than hard-coding them into the engine. In automation tools like Make.com, routing is determined by evaluating dynamic payloads against filters on connection edges, rather than relying on

fixed logic. TIVIDY needs interchangeable routing rules or sequential approval handlers that can evaluate dynamic payloads and determine the next step without altering the base engine classes.

A workflow system must react to external events and communicate with systems whose interfaces were not designed for it. When n8n interacts with an external API, it pauses execution and waits for a response. Also running work changes state over time (e.g., assigned, started, escalated), and operations valid at one point may be invalid at another. <https://www.uipath.com/community-blog/tutorials/getting-started-with-api-workflows-use-cases>  
{<https://flowwright.com/blog/api-integration-examples-guide>}

External communication must be wrapped in standard interfaces so the engine only interacts with generic objects not specific third-party APIs. The progression of a work item through its lifecycle requires state-specific behavior rather than manual type checks.

A real workflow system must explain what happened and preserve audit history. Platforms like Temporal record an append-only sequence of actions and state changes rather than just saving the current status.

Reporting and rollback operations should not clutter the core workflow model. TIVIDY must encapsulate individual actions as discrete objects that can be queued, executed, and reversed. Preserving snapshots of the workflow's internal state providing a standard way to traverse the active nodes will enable the system to generate analytics without exposing its internal structure.

## Task 2

### Scenario: Insurance Customer Complaint Handling

- The Organization: A financial services and insurance provider managing a high volume of customer support inquiries.
- The Process Being Managed: The intake, triaging, resolution, and documentation of customer complaints.
- Participants and Roles: The Customer (submits the complaint), the Triage Engine (automated router), the Domain Agent (e.g., Home Insurance Team member resolving the issue), and the Department Manager (handles escalated resolutions).
- Major Pieces of Work:
  1. External data intake via a web portal.
  2. Automated communication dispatch (confirmation emails).
  3. Database logging.
  4. Categorization and team assignment.
  5. Agent review and manual resolution action (e.g., drafting a response, issuing a refund).
- Important Decisions and Routing Points: A conditional routing point evaluates the insurance type (e.g., Homeowner vs. Auto) submitted in the payload to forward the task to the correct departmental inbox.

- Approvals and Dependencies: The "Agent Resolution" stage cannot begin until the "Intake and Database Logging" tasks are completed. Financial resolutions (like a refund) above a specific threshold require a dependency on a Manager's approval before the workflow can complete.
- Possible Failures or Exceptions: External database API downtime during the logging phase, missing mandatory payload data from the external portal, or a task violating its Service Level Agreement (SLA) time limit.
- External Systems: A REST API integration with a Google Sheets database for logging, and an external email/SMS gateway for automated customer follow-ups(bridge).
- Expected Outcome: The complaint is permanently logged, the customer receives a tailored resolution and follow-up, and the workflow instance archives itself as completed.

Basic concept:

